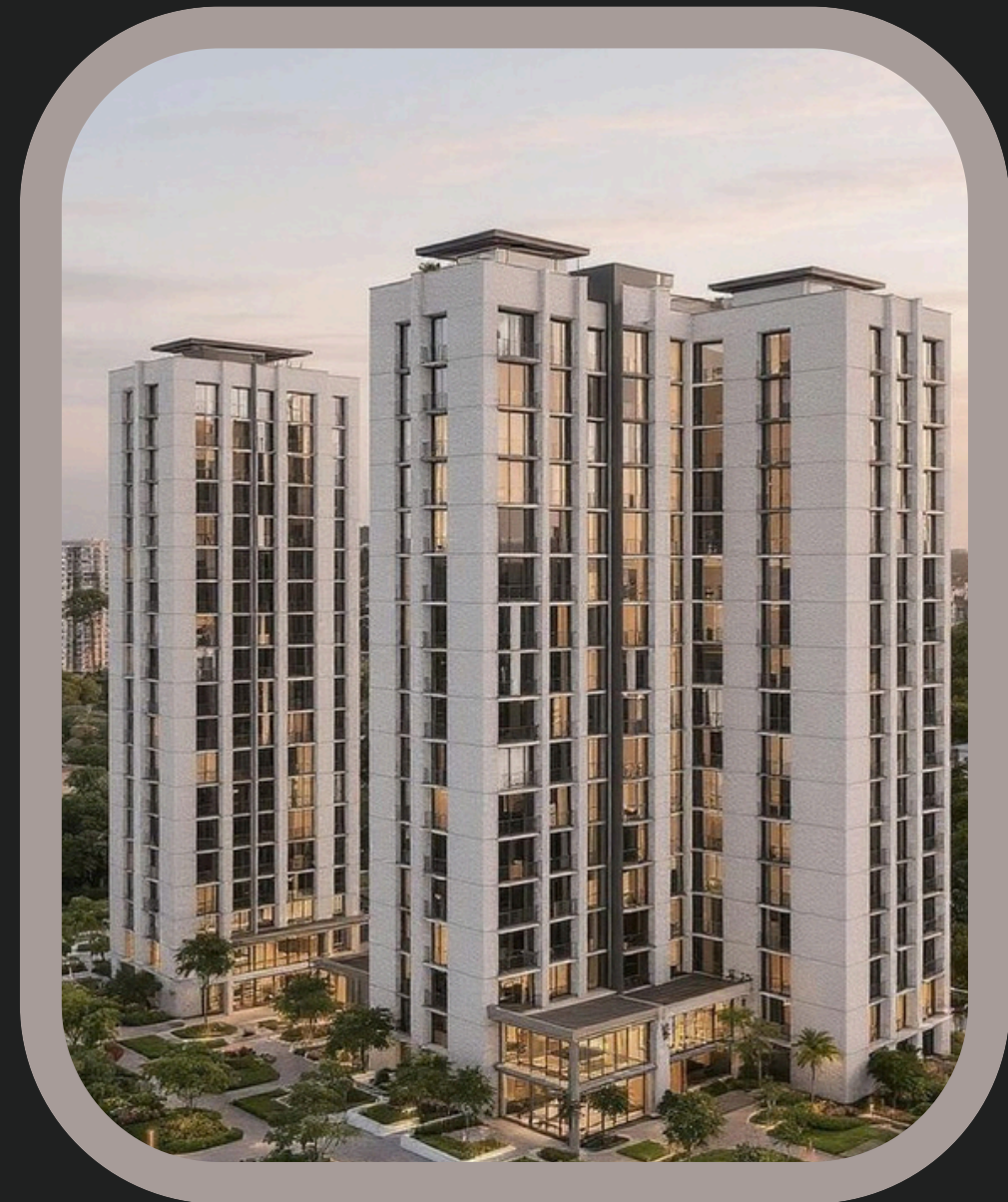
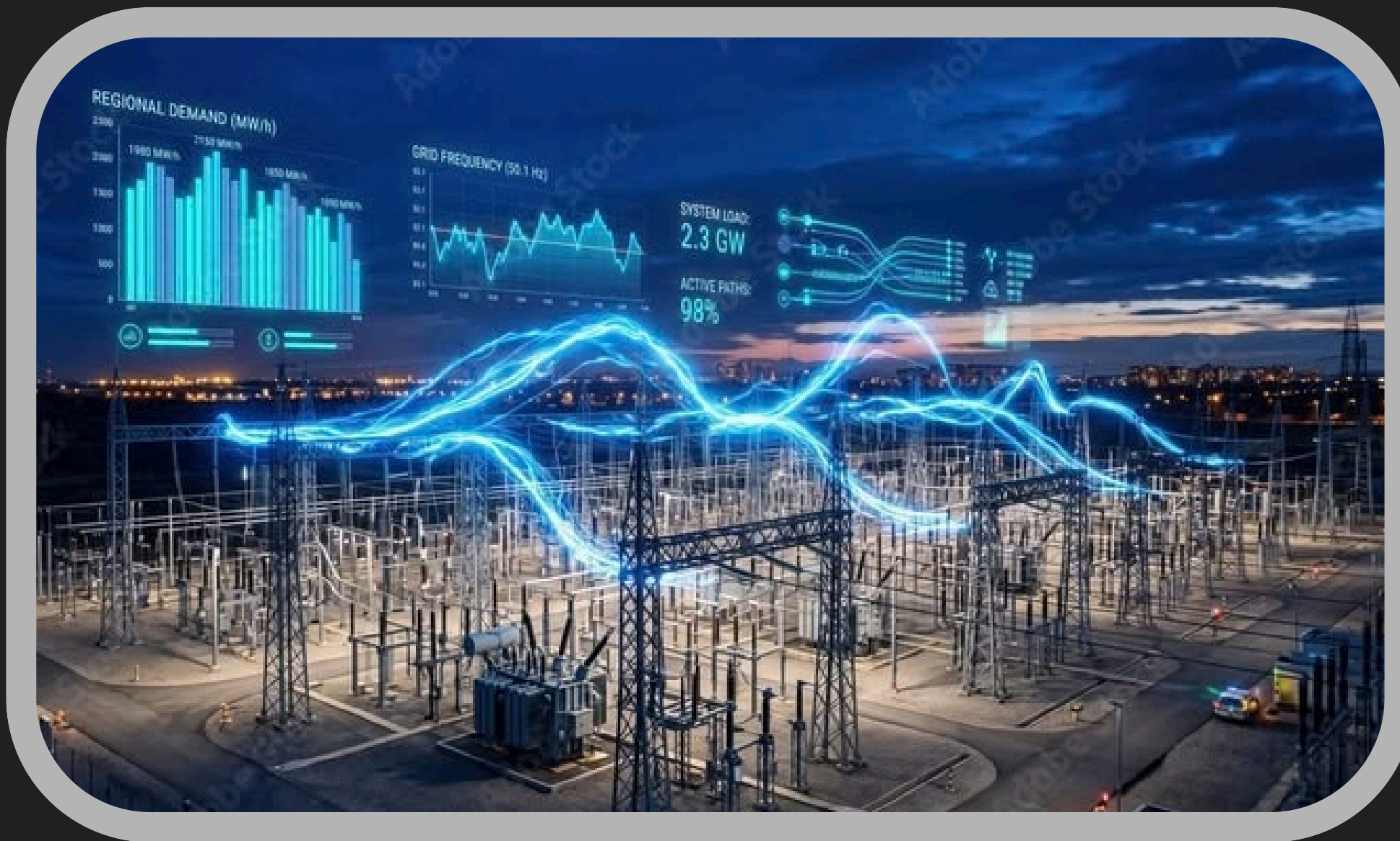
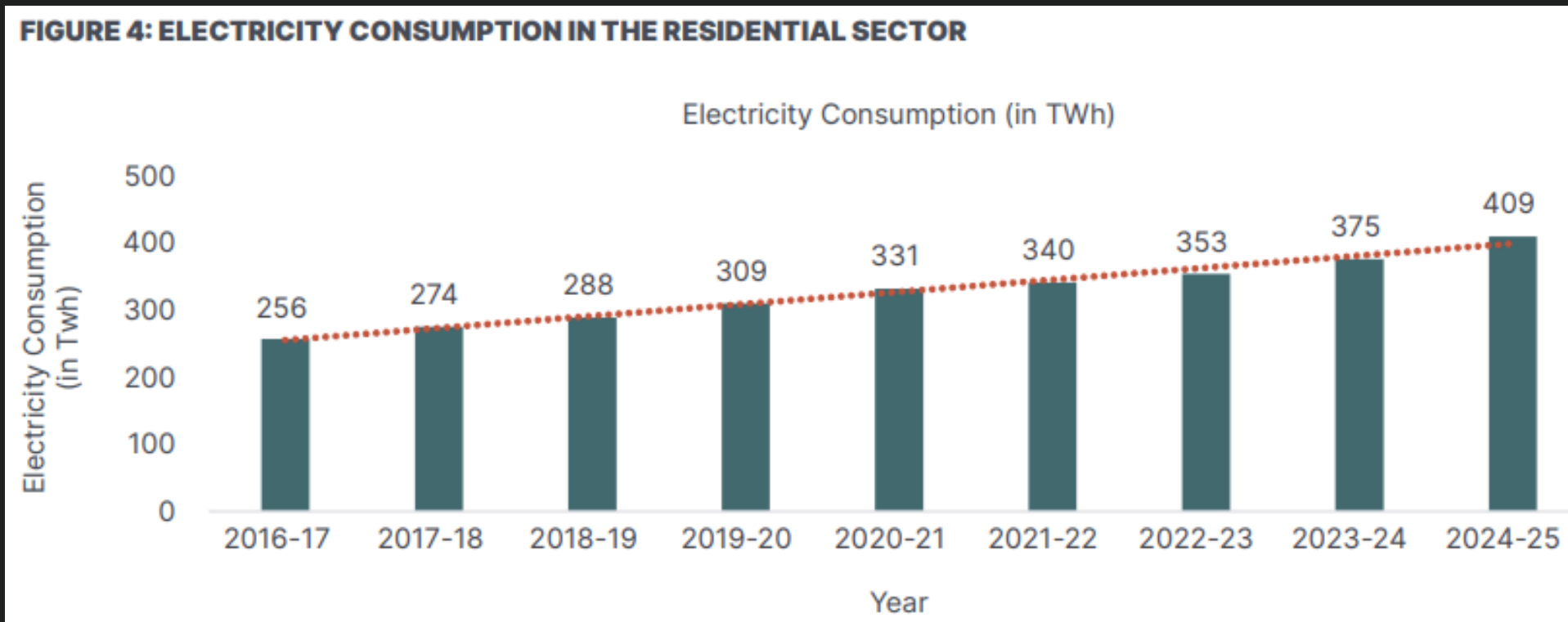


Gridantaw

A Residential Demand Response Platform that bridges residential communities and the grid during peak demand events.



Why Residential Communities?



- **25%** - Share of India's electricity consumption
- **256 → 409 TWh** - Growth in residential consumption
- **40%** - Electricity used by cooling loads

“Peak demand projected to surpass 300 GW by 2030”

“Residential communities represent a large, coordinated opportunity for demand response.”

6% Compound Annual Growth Rate (CAGR)
rise in residential electricity demand

Current Challenges

Residents	Building Managers	Grid / DISCOM
• Limited awareness of peak demand periods • Difficulty identifying flexible loads • Difficulty balancing comfort and savings	• Difficult to coordinate community-wide energy reduction. • Common loads and residential loads operate independently.	• Limited participation in demand response • No direct mechanism to communicate peak-demand actions to communities

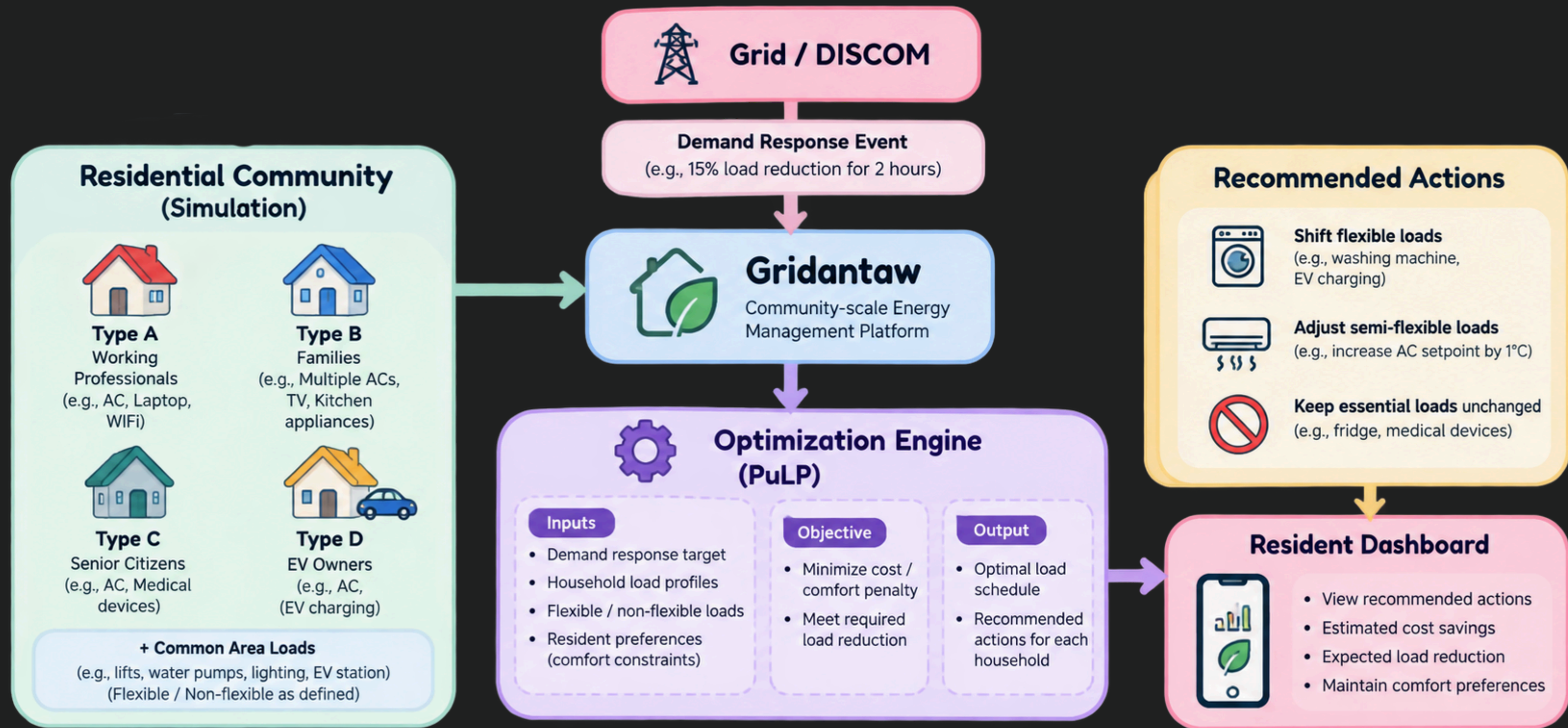
Introducing Gridantaw



As electricity demand continues to grow, peak demand periods place significant stress on the power grid. During these hours, electricity consumption can approach or exceed available supply, causing transmission lines, transformers, and substations to operate near their limits. If this stress is not managed effectively, equipment may overheat, relays would trip, and localized failures can cascade across the network, potentially resulting in large-scale blackouts.

Gridantaw aims to address this by connecting residential communities with demand response programs. During peak-demand events, residents receive timely alerts and personalized recommendations on which flexible loads can be shifted or reduced with minimal impact on comfort. The platform also provides clear estimates of potential cost savings thereby enabling informed participation while helping reduce stress on the grid.

Technical Approach



Technical Approach

The proposed solution will be demonstrated through a community-scale simulation of a residential complex. To model households with different energy consumption patterns and appliance usage, the Mesa agent-based framework will be used. Each household agent will represent different resident profiles, load categories, and preferences.

During a peak-demand event, a PuLP-based optimization engine will identify suitable load-shifting actions while also taking into account factors such as resident comfort, user preferences, and demand reduction targets. Finally, a user dashboard will provide residents with insights into their energy usage, estimated savings, and their contribution towards reducing community demand during peak periods.

USER JOURNEY

Step 1: Grid/DISCOM issues a peak-demand alert.

Step 2: Gridantaw identifies participating households and available flexible loads.

Step 3: Residents receive recommendations.

Step 4: User accepts or reviews recommendations.

Step 5: Resident sees estimated savings and contribution.

Step 6: Community demand reduces and peak stress on the grid decreases.

Key Features:



Real time
Peak Demand
Alerts

Community-
Level
Coordination

Savings
Visibility

Comfort-
Aware
Optimization

Expected Impact + Innovation

**Peak
Demand
Reduced**

**Resident
Comfort
Preserved**

**Improved
Grid
Reliability**

**Potential
Bill
Savings**

Alignment with Challenge

Challenge Need	Gridantaw Response
Energy Efficiency	Load shifting during peak demand
Smart Energy Management	Optimization using PuLP
User Engagement	Resident dashboard
Grid Reliability	Community demand reduction

Roadmap

Phase 1 → Community Simulation (Mesa)

Phase 2 → Optimization Engine (PuLP)

Phase 3 → Dashboard Development

Phase 4 → Integrated Prototype

Participant Introduction

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